frontiers | Frontiers in Chemistry

TYPE Original Research
PUBLISHED 13 January 2026
DOI 10.3389/fchem.2025.1750798

Check for updates

OPEN ACCESS

EDITED BY
Hai Lin,
University of Colorado Denver, United States

REVIEWED BY
Bolong Huang,
City University of Hong Kong, Hong Kong SAR, China
Rishu Khurana,
The University of Chicago, United States

*CORRESPONDENCE
Krishnan Thirumoorthy,
thirumoorthy.krishnan@vit.ac.in,
kthirumoorthy@gmail.com

RECEIVED 20 November 2025
REVISED 27 December 2025
ACCEPTED 29 December 2025
PUBLISHED 13 January 2026

CITATION
Malhan AH and Thirumoorthy K (2026) Catalytic hydrogenation of acetone to isopropyl alcohol with $CAl_3MgH_2^-$ containing planar tetracoordinate carbon.
Front. Chem. 13:1750798.
doi: 10.3389/fchem.2025.1750798

COPYRIGHT
© 2026 Malhan and Thirumoorthy. This is an open-access article distributed under the terms of the Creative Commons Attribution License (CC BY). The use, distribution or reproduction in other forums is permitted, provided the original author(s) and the copyright owner(s) are credited and that the original publication in this journal is cited, in accordance with accepted academic practice. No use, distribution or reproduction is permitted which does not comply with these terms.

Catalytic hydrogenation of acetone to isopropyl alcohol with $CAl_3MgH_2^-$ containing planar tetracoordinate carbon

Abdul Hamid Malhan[1] and Krishnan Thirumoorthy[1,2]*

[1]Department of Chemistry, School of Advanced Sciences, Vellore Institute of Technology, Vellore, Tamil Nadu, India, [2]School of Computer Science and Engineering, Vellore Institute of Technology, Vellore, Tamil Nadu, India

The current study, employing density functional theory, reports the hydrogenation of acetone to isopropyl alcohol catalyzed by $CAl_3MgH_2^-$, which contains a planar tetracoordinate carbon (ptC). Various computational approaches are employed to analyze acetone hydrogenation using the $CAl_3MgH_2^-$ as a potential catalyst. The reaction initiates with the carbonyl insertion into the Mg–H bond of the $CAl_3MgH_2^-$, followed by hydrogenation using molecular hydrogen (H_2). Analysis of natural atomic charges confirms that the H_2 molecule dissociates heterolytically into a proton–hydride pair, thereby regenerating the $CAl_3MgH_2^-$ in the product state. Intrinsic reaction coordinate calculations confirm the true connection between the reactant and product in the reaction pathway. This investigation highlights the potential of the ptC molecule as a catalyst and delineates the way for new opportunities in ptC-based catalysts.

KEYWORDS

acetone hydrogenation, $CAl_3MgH_2^-$, density functional theory, isopropyl alcohol, planar tetracoordinate carbon

Introduction

In recent years, the global market has experienced a surplus of acetone, primarily due to increased production from the cumene process used in phenol synthesis and bio-butanol fermentation (Papa, 2000; Weber and Weber, 2010). This increasing production of acetone, coupled with stable demand for its conventional derivatives like bisphenol A and methyl methacrylate, has led to a market surplus. This economic imbalance necessitates the valorization of excess acetone by converting it into more valuable chemicals. The catalytic hydrogenation of acetone to isopropyl alcohol is a promising and economically sustainable route to address this surplus while simultaneously meeting the rising demand for isopropyl alcohol (Mawarnis and Umar, 2019; Yang et al., 2017). The catalytic hydrogenation of acetone to synthesize isopropyl alcohol has been extensively investigated in both gas-phase (Al-Rabiah et al., 2022; Shen et al., 2021) and liquid-phase (Rahman and S-Al-Deyab, 2014). In gas-liquid hydrogenation reactions, the reaction rate can be limited by the mass transfer of the gaseous reactant (H_2) into the liquid phase. The low solubility of H_2 in organic solvents poses a significant challenge for liquid-phase hydrogenation, often requiring elevated pressures to achieve a sufficient dissolved hydrogen concentration at the catalyst surface. Conversely, gas-phase catalysis effectively circumvents this fundamental mass-transfer limitation. By vaporizing both reactants and passing them

over a catalyst, the system operates in a single phase, ensuring intimate contact among hydrogen, acetone, and the catalyst and thereby eliminating the gas-liquid interfacial barrier. The gas-phase catalytic hydrogenation of acetone to isopropyl alcohol is a strategically vital, environmentally benign chemical transformation that offers economic feasibility and attractiveness. Furthermore, the application of gas-phase catalysis offers substantial process intensification benefits compared to traditional liquid-phase systems, including the mitigation of mass-transfer limitations and the ability to operate at milder, near-ambient pressures. This leads to enhanced selectivity and simplified product separation, positioning it as a key technology for modern and sustainable chemical manufacturing. In 2022, Al-Rabiah et al. reported the gas-phase catalytic hydrogenation of acetone to isopropyl alcohol using a composite ruthenium on activated charcoal and nano-zinc oxide catalyst, achieving high selectivity (Al-Rabiah et al., 2022). The catalytic hydrogenation of carbonyl derivatives using molecular hydrogen is a vital process in modern synthetic organic chemistry and the production of both bulk and fine chemicals (Rylander, 1979). This type of hydrogenation typically involves well-defined noble metal complexes such as those of ruthenium, rhodium, or iridium (Chirila et al., 2024; Dub and Ikariya, 2012; Johnson et al., 2007; Noyori and Ohkuma, 2001). However, limited availability, substantial cost, and the toxicity of precious metals have sparked interest in developing more sustainable hydrogenation methods. As a result, there is a growing focus on identifying earth-abundant, main-group metals that can potentially replace these noble counterparts (Harder, 2012; Hill et al., 2016; Magre et al., 2022; Revunova and Nikonov, 2015; Roy et al., 2021). The development of well-defined main-group metal-based catalysts with comparable activity holds great promise for future industrial applications in chemistry.

In recent years, there has been a significant focus on magnesium-based catalysis, particularly since the pioneering structural characterization of Mg-hydride complexes (Bonyhady et al., 2010). Magnesium-based catalysts have shown promise in various catalytic reactions, including hydrosilylation (Garcia et al., 2019a; Garcia et al., 2019b; Rauch et al., 2017), hydroboration (Arrowsmith et al., 2012; Magre et al., 2019; Mukherjee et al., 2014; Szewczyk et al., 2019), hydroamination (Dunne et al., 2010; Zhang et al., 2012), hydrogenation (Liang et al., 2022), and dehydrocoupling (Liptrot et al., 2014; Ried et al., 2019; Wirtz et al., 2022) marking considerable progress in this field. The magnesium-based catalyst has proven to be significant in asymmetric catalysis (Wang et al., 2024; Yang et al., 2019), with applications in asymmetric thia-Michael addition (Jaszczewska-Adamczak et al., 2024), asymmetric hydroalkylation (Ye et al., 2025), and asymmetric hydroboration (Falconnet et al., 2019). One of the primary challenges in using main-group metal compounds as catalysts is their limited ability to activate H_2 bonds and their high reactivity, which often leads to unexpected side reactions. Therefore, it is crucial to develop stable main-group metal complexes with enhanced catalytic activity to address these issues. Both experimental and theoretical studies have highlighted stable main-group molecules with planar hypercoordination (Das et al., 2023; Inostroza et al., 2023; Leyva-Parra et al., 2021; Sun et al., 2020; Sun et al., 2022; Wang et al., 2023; Yang et al., 2015; Zhang et al., 2021), offering a potential solution. The study of planar tetracoordinate carbon (ptC) has attracted significant interest since its initial discovery in 1968 by H. J. Monkhorst (Monkhorst, 1968). Experimentally identified species such as CAl_4^{2-}, CAl_4^-, and $C_2Al_5^-$ containing ptC have been reported (Li et al., 1999; 2000; Tsuruoka et al., 2018; Zhang et al., 2023). Additionally, $C_2Al_4^-$ and $C_5Al_5^-$, containing ptC, were initially computationally predicted, subsequently leading to their experimental identification (Naumkin, 2008; Wu et al., 2011; Zhang et al., 2021; Zhang et al., 2023). The global minimum ptC of $CAl_4H^{0/-}$ containing hydrogen was both experimentally and theoretically reported (Xu et al., 2017). Additionally, planar hypercoordinate systems of $CAl_nBe_mH_x^q$ ($n + m = 5$, $q = 0, \pm1$, $x = q + m - 1$), $Al_2C_4H_2$, $CBe_2H_5^-$ and $Si_2C_5H_2$ containing hydrogen have also been reported (Jin et al., 2024; Malhan et al., 2022; Thirumoorthy et al., 2020; Zhao et al., 2018). The application of planar hypercoordinate carbon systems for reversible hydrogen storage properties was reported in 2023 (Sarmah et al., 2023). Notably, it was found that the planar hypercoordinate carbon systems of CSi_2Li_2, $CBe_5Li_5^+$, and $CS_3Li_3^+$ exhibit high gravimetric densities of 25.6, 39.7, and 13.5 wt% for hydrogen storage. Additionally, the research suggested potential applications of planar hypercoordinate systems in optoelectronics, electronics, and photovoltaics (Yang et al., 2015). In 2025, Zhang et al. (2025) computationally reported the ptC catalysts $C_2B_2Me_2$ and $C_2B_2tBu_2$ in thermodynamically stable Au^I complexes, which were examined for their catalytic activity in the allylic acetate rearrangement. It demonstrated that the ptC catalyst $(C_2B_2Me_2)Au^I$ significantly outperforms the conventional N-heterocyclic carbene–Au^I system in the allylic acetate rearrangement. This superior activity stems from a considerably lower energy barrier, indicating that the reaction is both kinetically and thermodynamically more favorable with the ptC catalyst. This enhanced performance is directly related to the ptC's unique electronic structure, which establishes a more beneficial electronic environment for the Au center throughout the catalytic cycle. Significantly, the inherent planarity and electron delocalization of the ptC function as active contributors rather than mere passive characteristics, thus driving the observed increase in catalytic activity.

Our research on the earth-abundant main-group elements has revealed that ptC is a global minimum structure in $CAl_3MgH_2^-$ in its singlet state (Malhan and Thirumoorthy, 2024). Extensive analysis using various quantum chemical tools has confirmed the stability of this structure. The study identified that the terminal hydrogen (−0.71 |e|) attached to magnesium exhibits a higher hydridic nature compared to the bridging hydrogen (−0.40 |e|). The bridging hydrogen, due to its overlap with two aluminum atoms, requires more energy to break a bond, rendering it less reactive than the readily available terminal hydrogen, which has a higher hydridic character. This discovery of hydridic hydrogen in $CAl_3MgH_2^-$ has sparked our interest in exploring its potential as a catalyst, thereby expanding the application of ptCs in hydrogenation reactions. Moreover, a recent theoretical study by our research group (Malhan and Thirumoorthy, 2025) reported that $CAl_3MgH_2^-$ can catalyze alkyne hydrogenation, underscoring the need for further investigation into its catalytic properties and mechanistic pathways.

Despite previous and ongoing experimental efforts involving ptC, there is a lack of reliable laboratory techniques for synthesizing stable ptC that can be isolated, investigated, and handled under

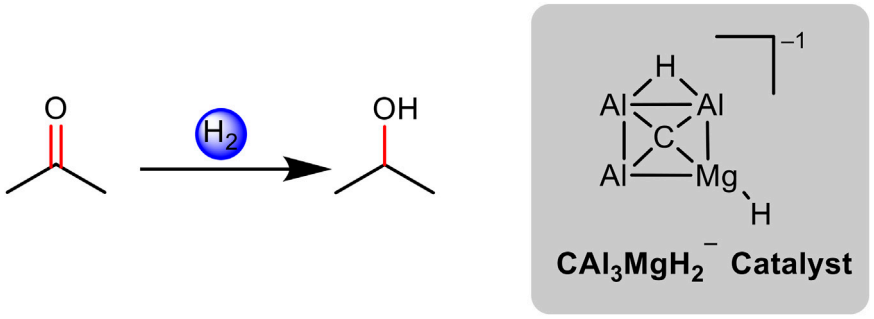


FIGURE 1
Acetone hydrogenation using $CAl_3MgH_2^-$ catalyst.

ambient conditions. However, an in-depth exploration of ptC structures could facilitate the development of novel materials with various chemical and physical properties, paving the way for new research areas.

Given the extensive research and established knowledge in ptC chemistry, viable synthetic methodologies for ptC are anticipated to materialize in the future. Consequently, the importance of the $CAl_3MgH_2^-$, which has demonstrated stability, is significantly enhanced, particularly for its potential as a catalyst in hydrogenation reactions. We have addressed the stability of $CAl_3MgH_2^-$ in the previous work (Malhan and Thirumoorthy, 2024), the current study reports acetone hydrogenation using $CAl_3MgH_2^-$ at room temperature (298.15 K) and 1 atm pressure (Figure 1), supported by quantum chemical calculations.

Computational methodology

Density Functional Theory (DFT) was utilized to study the hydrogenation of acetone. The geometry optimization was carried out at the ωB97XD/def2-TZVPP (Chai and Head-Gordon, 2008; Weigend and Ahlrichs, 2005; Zheng et al., 2011) level, and harmonic vibrational frequency analysis was performed at the same level to confirm whether the structures represent minima or transition states on the potential energy surface. The ωB97XD functional incorporates a long-range correction term to enhance the description of van der Waals interactions, particularly for noncovalent interactions, dispersion forces, and molecular properties. The def2-TZVPP basis set comprises a triple-zeta valence with polarization functions, providing a detailed electron description and effectively capturing non-covalent interactions. Additionally, all optimized geometries at the ωB97XD/def2-TZVPP level underwent optimization, and harmonic frequency analysis at meta-hybrid M06/def2-TZVPP level (Weigend and Ahlrichs, 2005; Zhao and Truhlar, 2008; Zheng et al., 2011) to ensure that the energy trends were consistent with the results obtained using the former method. To further validate the energetics obtained from the ωB97XD/def2-TZVPP level, single-point energy calculations were subsequently conducted using the DLPNO-CCSD(T)/def2-TZVPP (Riplinger et al., 2013; Riplinger and Neese, 2013) method on the optimized geometries of the ωB97XD/def2-TZVPP level, which was performed using the ORCA program (Neese, 2012). The transition state structures were identified using the Berny algorithm (Schlegel, 1982) within the Gaussian 16 package, which is designed to optimize for a transition state rather than a local minimum. This is achieved through a quasi-Newton method that approximates the Hessian (the second derivative matrix). This method enables the identification of a stationary point on the potential energy surface, specifically the first-order saddle point (transition state). This critical point is characterized by a Hessian matrix possessing exactly one negative eigenvalue, which corresponds to the reaction coordinate. The optimization protocol is rigorously managed to maintain this single negative eigenvalue throughout the search. Essentially, the algorithm proceeds uphill to locate the transition state, which corresponds to a maximum in one direction and a minimum in all others. Various quantum chemical computational tools were employed to comprehensively characterize the chemical bonding properties along the entire reaction pathway. Intrinsic reaction coordinate (IRC) calculations (Gonzalez and Schlegel, 1989) were performed to determine the minimum-energy reaction pathway connecting the dynamics of a reaction. Furthermore, natural population analysis (Reed et al., 1985) within the natural bond orbital framework (Nobel et al., 2017) was performed using the Gaussian 16 program (Frisch et al., 2016) to obtain natural atomic charges. Moreover, non-covalent interaction (NCI) analysis was performed using Multiwfn 3.8 program (Lu and Chen, 2012). The analyses as mentioned above were carried out at the ωB97XD/def2-TZVPP level, and all the calculations, including optimization, frequency, and IRC, were performed with the Gaussian 16 package (Frisch et al., 2016).

Results and discussion

The stability of $CAl_3MgH_2^-$ has been confirmed through computational studies in the previous study, which emphasized the delocalization of electron density and the presence of double aromaticity (Malhan and Thirumoorthy, 2024). A similar isoelectronic structure, CAl_4H^- (Xu et al., 2017), has been experimentally identified, indicating that substituting one aluminum atom with an isoelectronic Mg–H unit may facilitate the identification of $CAl_3MgH_2^-$.

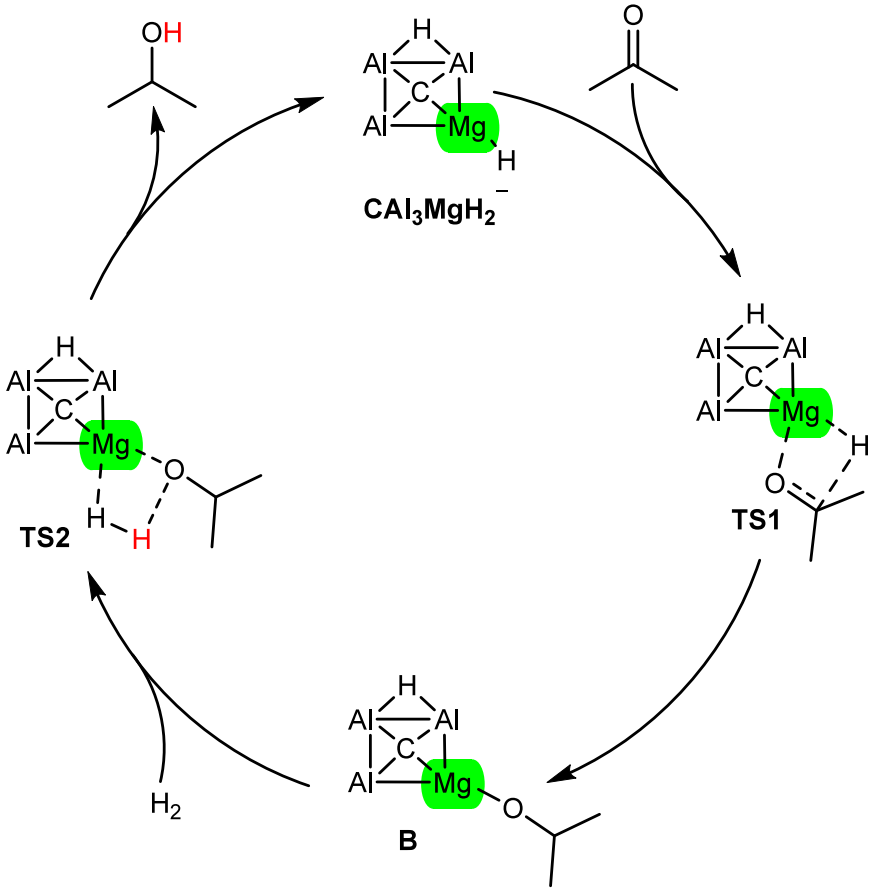


FIGURE 2
Proposed mechanism for acetone hydrogenation using the $CAl_3MgH_2^-$.

The proposed mechanism of acetone hydrogenation using the $CAl_3MgH_2^-$ is illustrated in Figure 2. The reaction begins with the coordination of the carbonyl (C═O) to the Mg–H bond. This interaction facilitates the transfer of a hydride ion (H^-) from the Mg–H bond to the carbonyl carbon of acetone, proceeding through the transition state TS1. This step results in the formation of an Mg–O bond and produces intermediate B. Subsequently, the introduction of H_2 leads to the formation of another transition state, TS2. In this step, the H_2 molecule undergoes cleavage, with one hydrogen atom added to the magnesium atom of the catalyst and another to the oxygen atom. This generates the desired product, isopropyl alcohol, while simultaneously regenerating the $CAl_3MgH_2^-$. The present work provides a detailed investigation into the hydrogenation of acetone facilitated by the $CAl_3MgH_2^-$, providing valuable insights into the mechanisms involved and the overall efficiency of the catalytic process.

The Gibbs free energy profile depicted in Figure 3 illustrates the hydrogenation of acetone using the $CAl_3MgH_2^-$ under standard computational conditions (298.15 K and 1 atm pressure), calculated at the ωB97XD/def2-TZVPP level of theory. Additionally, their zero-point corrected energy profile is displayed in Supplementary Figure S1. Their ZPVE corrections, and the number of imaginary frequencies of all optimized stationary points involved in the hydrogenation of acetone using $CAl_3MgH_2^-$ at the ωB97XD/def2-TZVPP and M06/def2-TZVPP levels are listed in Supplementary Table S1 and Supplementary Table S2, respectively. Both density functional methods, transition states TS1 and TS2, each exhibit one negative imaginary frequency, confirming their identity as first-order saddle points on the potential energy surface. Conversely, structures A, B, C, and D possess zero imaginary frequencies, indicating they correspond to minima along the reaction coordinate. The reaction is initiated by the coordination of the carbonyl (C═O) to the Mg–H bond of the $CAl_3MgH_2^-$, in reactant state A. Subsequently, carbonyl (C═O) is inserted into the Mg–H bond with an energy barrier of 6.96 kcal/mol, forming a stable intermediate B with −30.06 kcal/mol. Following this, the hydrogenation of the Mg–O bond with molecular hydrogen (H_2) leads to heterolytic H_2 cleavage via a four-center transition state (TS2) with an energy barrier of 21.59 kcal/mol. This step, which regenerates the $CAl_3MgH_2^-$ and forms the desired product, isopropyl alcohol, in product state D, is the slowest in the reaction. The energy values obtained at the ωB97XD/def2-TZVPP level exhibit a notable similarity with the energy trends observed at the M06/def2-TZVPP level. A comparison of their respective Gibbs free energy and Zero-point corrected energy values is given in Supplementary Table S3. Specifically, the energy values for the transition states (TS1 and TS2) at the M06/def2-TZVPP level indicate a lower activation energy barrier compared to those at the ωB97XD/def2-TZVPP level.

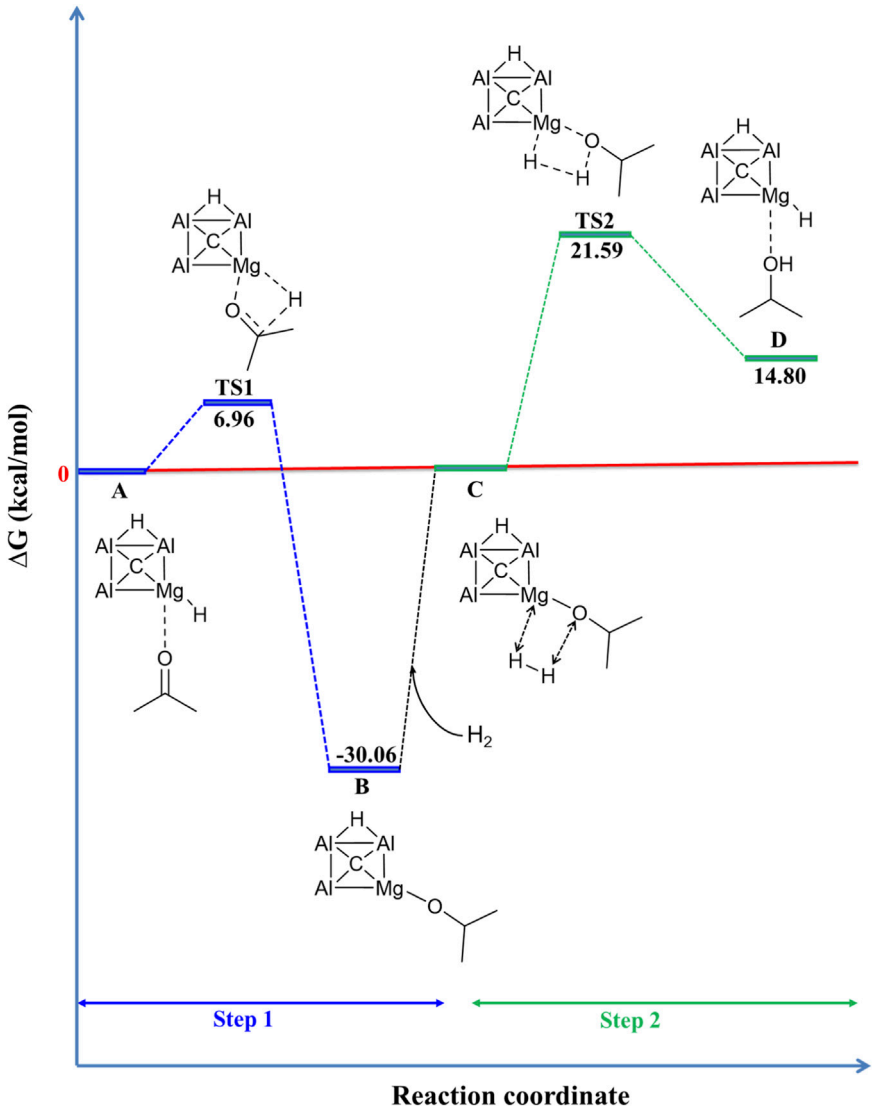


FIGURE 3
Gibbs free energy profile in kcal/mol for hydrogenation of acetone using $CAl_3MgH_2^-$ at the ωB97XD/def2-TZVPP level. The reaction involves two distinct transition states, each with an activation barrier of 6.96 kcal/mol and 21.59 kcal/mol, respectively.

TABLE 1 Comparison of ΔE calculated at the ωB97XD/def2-TZVPP and CCSD(T)/def2-TZVPP//ωB97XD/def2-TZVPP levels for hydrogenation of acetone using $CAl_3MgH_2^-$.

Structure labels	Energy (kcal/mol)	
	ωB97XD/def2-TZVPP	DLPNO-CCSD(T)/def2-TZVPP//ωB97XD/def2-TZVPP
	ΔE	ΔE
A	0.00	0.00
TS1	6.58	5.89
B	−32.94	−34.76
C	0.00	0.00
TS2	17.95	17.92
D	8.47	8.15

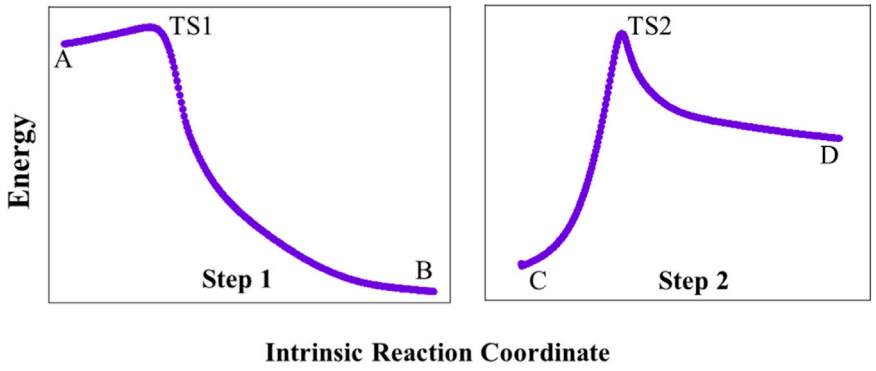


FIGURE 4
Intrinsic reaction coordinate pathway for hydrogenation of acetone using $CAl_3MgH_2^-$. Both the transition states are truly connected to their adjacent local minima. Intrinsic reaction coordinate calculations are performed at the ωB97XD/def2-TZVPP level (Intrinsic reaction coordinate video is provided in the Supplementary Material).

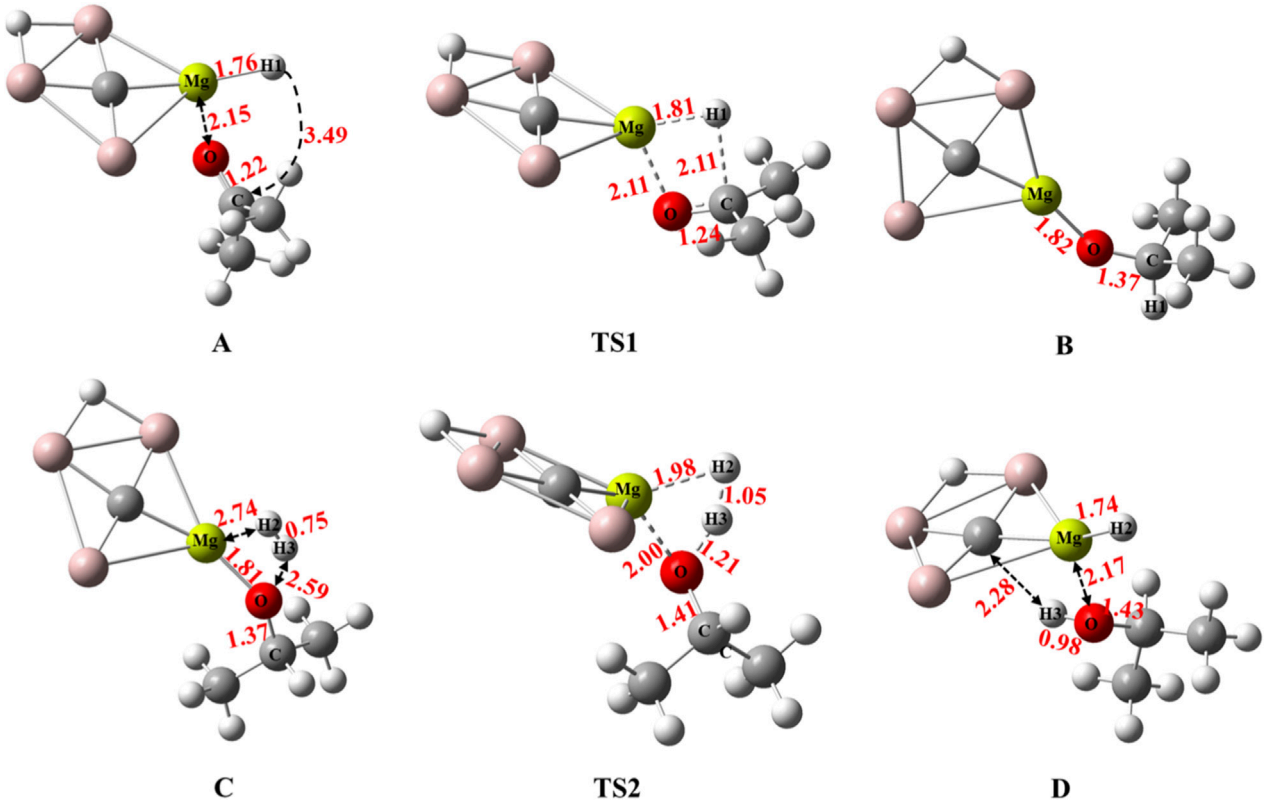


FIGURE 5
Calculated bond lengths in Å for optimized structures of the stationary points along the reaction pathway for hydrogenation of acetone using $CAl_3MgH_2^-$ at the ωB97XD/def2-TZVPP level. The elongation of the Mg–H1 bond in TS1 confirms the transfer of H1. In TS2, the elongation of H_2 confirms its cleavage.

However, the energy values across both methodologies remain highly comparable. This alignment highlights the methodological consistency between these two distinct functionals in assessing the reaction energy profile. Such marked consistency across these two different functionals not only enhances the robustness of the computational predictions but also reinforces the credibility of the methodologies employed in this study. Furthermore, the comparison of ΔE values calculated at ωB97XD/def2-TZVPP and CCSD(T)/def2-TZVPP//ωB97XD/def2-TZVPP levels is given in Table 1. The results confirm that the reaction energy trends remain consistent and comparable between the two methods. This further provides strong evidence that the ωB97XD/def2-

TABLE 2 Natural atomic charges (|e|) on atoms involved in the reaction pathway of hydrogenation of acetone using $CAl_3MgH_2^-$ calculated at the ωB97XD/def2-TZVPP and M06/def2-TZVPP levels. (Atom labels are followed as given in Figure 5).

Structure labels	Methods	Mg	O	C	H1	H2	H3
A	ωB97XD/def2-TZVPP	1.58	−0.63	0.64	−0.74	—	—
	M06/def2-TZVPP	1.58	−0.65	0.64	−0.74	—	—
TS1	ωB97XD/def2-TZVPP	1.64	−0.75	0.62	−0.65	—	—
	M06/def2-TZVPP	1.63	−0.76	0.64	−0.64	—	—
B	ωB97XD/def2-TZVPP	1.75	−1.10	0.13	0.12	—	—
	M06/def2-TZVPP	1.74	−1.10	0.14	0.11	—	—
TS2	ωB97XD/def2-TZVPP	1.67	−0.94	0.09	0.16	−0.45	0.29
	M06/def2-TZVPP	1.65	−0.93	0.10	0.15	−0.48	0.31
D	ωB97XD/def2-TZVPP	1.57	−0.83	0.08	0.18	−0.71	0.54
	M06/def2-TZVPP	1.56	−0.84	0.08	0.18	−0.70	0.54

TZVPP method describes the reaction energetics with near accuracy.

The identification of transition states was achieved through the Berny algorithm implemented in the Gaussian 16 package (Frisch et al., 2016), with a specific focus on the hydrogenation process of acetone catalyzed by $CAl_3MgH_2^-$. To gain deeper insights into reaction dynamics, IRC calculations were employed, enabling a comprehensive exploration of the minimum-energy pathways associated with the chemical reaction. These calculations are crucial for elucidating the reaction pathway, providing a comprehensive overview of how reactants are systematically converted into products. The IRC calculations were performed using the transition states TS1 and TS2 identified by the Berny algorithm, and their IRC pathways are given in Figure 4. This analysis confirmed that these transition states are strongly interconnected with both the reactants and the products, thus reinforcing the overall integrity of the proposed reaction pathway. The findings demonstrate that the calculated pathway is energetically feasible, ensuring a reliable depiction of the hydrogenation process.

The optimized geometries on the potential energy surface, including bond lengths, for the hydrogenation of acetone using the $CAl_3MgH_2^-$, are given in Figure 5 along with their corresponding natural atomic charges from natural population analysis as provided in Table 2 at the ωB97XD/def2-TZVPP level. Additionally, the optimized geometries are displayed in Supplementary Figure S2 at the M06/def2-TZVPP level. In reactant state A, the hydrogen from the Mg–H1 bond has a natural atomic charge of −0.74 |e|, which indicates its hydridic nature, and the Mg–H1 bond distance is 1.76 Å. As the reaction proceeds, in TS1 the Mg–H1 and C═O bond elongate to 1.81 Å and 1.24 Å, respectively. This elongation facilitates the H1 hydride shift to the carbonyl carbon and the formation of the Mg–O bond. As a result, intermediate B is formed, where the Mg–O bond measures 1.82 Å, and the carbonyl (C═O) reduces into a single C–O bond with a bond length of 1.37 Å. The natural atomic charges on oxygen (−1.10 |e|) and magnesium (1.75 |e|) indicate that they now act as Lewis base and Lewis acid, respectively. This Lewis acid–base pair dissociates H_2 molecules heterolytically into a proton–hydride pair, thereby favoring the hydrogenation process (Aireddy and Ding, 2022). The natural atomic charges of H2 (−0.45 |e|) and H3 (0.29 |e|) in TS2, in conjunction with the elongation of the H2–H3 bond from 0.75 Å to 1.05 Å, confirm the heterolytic cleavage of H_2, which leads to the formation of $Mg–H2^{\delta-}$ and $O–H3^{\delta+}$ pairs. This process results in regenerating the $CAl_3MgH_2^-$, and the formation of the desired product, isopropyl alcohol in D. To verify the complete regeneration of the $CAl_3MgH_2^-$, the bonding parameters of the $CAl_3MgH_2^-$ within the product complex D are compared with those of the isolated $CAl_3MgH_2^-$, as shown in Supplementary Figure S3. The results indicate that all bond lengths in the $CAl_3MgH_2^-$ of product complex D are identical to those of the isolated $CAl_3MgH_2^-$, with no discernible changes. This consistent bond-length measurement provides compelling evidence that the $CAl_3MgH_2^-$ has been fully regenerated, thereby preserving its structural integrity. Additionally, a comparison of bond lengths for all optimized structures along the reaction pathway for the hydrogenation of acetone using the $CAl_3MgH_2^-$, calculated at the ωB97XD/def2-TZVPP and M06/def2-TZVPP levels, is provided in Supplementary Table S4. Remarkably, the bond lengths calculated from both levels of theory show a high degree of similarity. This consistency in the results reinforces the reliability of the findings across the two theoretical frameworks employed in this study.

The functional significance of the ptC resides in its unique electronic configuration, which confers absolute planarity even within the transition states. As Figure 5 illustrates, the structural integrity of the ptC core is maintained throughout the reaction. This structural integrity is not merely geometric but stems directly from the extensive electron delocalization in the planar $CAl_3MgH_2^-$ core. Critically, this delocalization prevents distortion of the $CAl_3MgH_2^-$ core throughout the reaction. Furthermore, the central ptC facilitates catalytic activity by charge stabilization, effectively compensating for charge perturbations at the magnesium active site. This is supported by the natural atomic charges (Table 2), which exhibit minimal variation in the magnesium natural charges throughout the reaction, confirming that the ptC core efficiently prevents destabilizing charge accumulation and effectively stabilizes the transition state.

The NCI analysis was performed using converged Self-Consistent Field (SCF) calculations with a cutoff value of 0.5 to elucidate the nature of interactions governing the hydrogenation of acetone using the $CAl_3MgH_2^-$. NCI provides insights into van der Waals interactions, hydrogen bonds, π–π stacking, and other weak interactions, offering a visual representation of non-covalent interactions in a molecular system. The 3D isosurfaces and 2D reduced density gradient (RDG) graphs for all the stationary points along the reaction pathway were generated and are shown in Figure 6, explicitly illustrating the interactions pertinent to bond

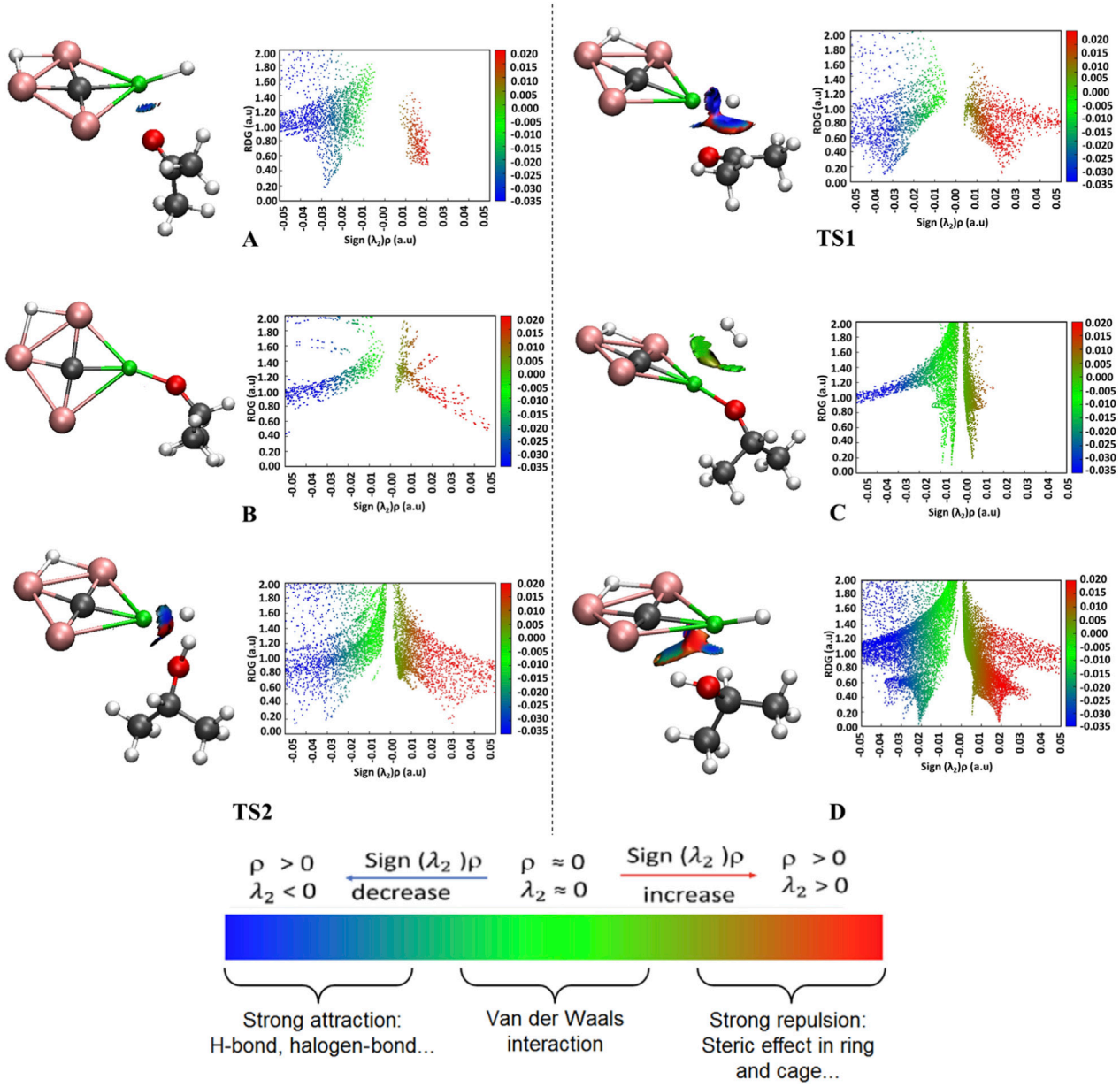


FIGURE 6
NCI, 3D isosurfaces (on left) and 2D-RDG graphs (on right) for the specific interactions associated with bond breaking and formation involved in the reaction pathway of hydrogenation of acetone using $CAl_3MgH_2^-$ at ωB97XD/def2-TZVPP level. The reaction initiates with van der Waals interactions (green isosurface) that bring the reactants together in the initial complex. As the system approaches the transition state, these evolve into strong, electrostatic interactions (blue isosurface), which are critical for product formation. Isosurfaces are colored as: strong attraction (blue), van der Waals interaction (green), and repulsive interaction (red). The complete 3D isosurfaces and their corresponding RDG graphs are given in Supplementary Figure S4.

cleavage and formation. For a comprehensive perspective, the complete 3D isosurfaces and their corresponding RDG graphs are given in Supplementary Figure S4. The atom labels used here follow those in Figure 5. In reactant state A, the presence of blue patches between the acetone and the $CAl_3MgH_2^-$ signifies an electrostatic attraction, which initiates the reaction process. As the reaction proceeds in TS1, an increase in blue patches between magnesium and oxygen atoms, as well as between the carbonyl carbon and H1 hydrogen atoms, was observed. These patches indicate a strong electrostatic attraction between the atoms, which facilitates the bond formation process leading to intermediate B. Following the bond formation process in B, no non-covalent interactions are observed, confirming the chemical bond between Mg–O and C–H1. As molecular hydrogen is introduced into C, the presence of green patches between the Mg and H_2 was observed, indicating van der Waals interactions. This interaction leads to the transition state TS2, where H_2 is cleaved and added to Magnesium and oxygen. Specifically, the bonding interaction between the O–H3 and between the Mg–H2 enhances the H_2 heterolytic cleavage, leading to product state D. The presence of blue patches between the Mg and O in product state D indicates a strong attraction between them. Additionally, blue patches between

H3 (0.54 |e|) and the central carbon (−2.76 |e|) of $CAl_3MgH_2^-$ were observed with a 2.28 Å distance between them, indicating H-bonding. These attractive forces serve as stabilizing factors for product state D, lowering its energy compared to the combined energies of the individual product and $CAl_3MgH_2^-$.

To investigate the impact of the $CAl_3MgH_2^-$, a comprehensive analysis of the reaction pathway for the hydrogenation of acetone was performed, comparing scenarios both with and without the $CAl_3MgH_2^-$. Supplementary Figure S5 illustrates the energy barriers associated with the transition states during this hydrogenation reaction. Additionally, comparison with the zero-point corrected energy profile is given in Supplementary Figure S6**.** In the absence of a $CAl_3MgH_2^-$, the energy barrier for reaching the transition state is relatively high, at 73.82 kcal/mol. In contrast, introducing the $CAl_3MgH_2^-$ significantly lowers the energy barriers, revealing a more favorable reaction pathway. Notably, using the $CAl_3MgH_2^-$ results in two distinct transition-state energy barriers: the lower of the two, at 6.96 kcal/mol, indicating a markedly reduced activation energy requirement. The second transition state, where the hydrogenation occurs, is at 21.59 kcal/mol. These considerable activation energy barriers demonstrate the effectiveness of the $CAl_3MgH_2^-$, which decreases the activation energy barrier for the hydrogenation step by a notable 52.23 kcal/mol compared to the case without a $CAl_3MgH_2^-$. Such a substantial drop not only enhances the feasibility of the hydrogenation process but also emphasizes the $CAl_3MgH_2^-$ role in facilitating the reaction under more energetically favorable conditions.

Conclusion

In summary, this study investigates the catalytic properties of the ptC molecule, particularly in acetone hydrogenation. Utilizing computational approaches, the present work systematically examined the catalytic activity of $CAl_3MgH_2^-$ in the hydrogenation of acetone to isopropyl alcohol. The reaction mechanism commences with the carbonyl group of acetone inserted into the magnesium-hydride (Mg–H) bond of the $CAl_3MgH_2^-$. Following this step, the reaction progresses through a heterolytic dissociation of the H_2 molecule into a proton-hydride pair. This key transitional stage is essential, as it drives the hydrogenation reaction and facilitates $CAl_3MgH_2^-$ regeneration, ultimately producing the desired isopropyl alcohol. Furthermore, the IRC calculations confirmed that the reaction mechanism follows a minimum-energy pathway, demonstrating strong interconnections between the transition states, reactant, and product. The heterolytic dissociation of H_2, assisted by the Lewis acid-base pair, is supported by natural atomic charges. The findings underscore the $CAl_3MgH_2^-$, demonstrating its ability for acetone hydrogenation, highlighting it as a potential catalyst for hydrogenation reactions. Although several ptC have been observed in the gas phase, effective laboratory synthesis methods are still lacking. Nevertheless, ongoing research in this area is expected to lead to viable synthesis approaches in the future.

Data availability statement

The original contributions presented in the study are included in the article/Supplementary Material, further inquiries can be directed to the corresponding author.

Author contributions

AHM: Conceptualization, Data curation, Formal Analysis, Investigation, Methodology, Validation, Visualization, Writing – original draft. KT: Conceptualization, Data curation, Formal Analysis, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – review and editing.

Funding

The author(s) declared that financial support was received for this work and/or its publication. The authors express gratitude to the management of the Vellore Institute of Technology for providing the necessary facilities and covering the article processing charge for this manuscript.

Acknowledgements

The computational facility at VIT, Vellore, India, used for this work is gratefully acknowledged.

Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that generative AI was not used in the creation of this manuscript.

Any alternative text (alt text) provided alongside figures in this article has been generated by Frontiers with the support of artificial intelligence and reasonable efforts have been made to ensure accuracy, including review by the authors wherever possible. If you identify any issues, please contact us.

Publisher's note

All claims expressed in this article are solely those of the authors and do not necessarily represent those of their affiliated organizations, or those of the publisher, the editors and the reviewers. Any product that may be evaluated in this article, or claim that may be made by its manufacturer, is not guaranteed or endorsed by the publisher.

Supplementary material

The Supplementary Material for this article can be found online at: https://www.frontiersin.org/articles/10.3389/fchem.2025.1750798/full#supplementary-material

References

Aireddy, D. R., and Ding, K. (2022). Heterolytic dissociation of H_2 in heterogeneous catalysis. *ACS Catal.* 12, 4707–4723. doi:10.1021/acscatal.2c00584

Al-Rabiah, A. A., Boz, I., Akhmedov, V. M., Mostafa, M. M. M., and Bagabas, A. A. (2022). Highly selective gas-phase catalytic hydrogenation of acetone to isopropyl alcohol. *Catalysts* 12, 1251. doi:10.3390/catal12101251

Arrowsmith, M., Hadlington, T. J., Hill, M. S., and Kociok-Köhn, G. (2012). Magnesium-catalysed hydroboration of aldehydes and ketones. *Chem. Commun.* 48, 4567–4569. doi:10.1039/c2cc30565h

Bonyhady, S. J., Jones, C., Nembenna, S., Stasch, A., Edwards, A. J., and McIntyre, G. J. (2010). β-Diketiminate-Stabilized Magnesium(I) dimers and Magnesium(II) hydride complexes: synthesis, characterization, adduct formation, and reactivity studies. *Chem. - Eur. J.* 16, 938–955. doi:10.1002/chem.200902425

Chai, J.-D., and Head-Gordon, M. (2008). Long-range corrected hybrid density functionals with damped atom–atom dispersion corrections. *Phys. Chem. Chem. Phys.* 10, 6615–6620. doi:10.1039/b810189b

Chirila, A., Hu, Y., Linehan, J. C., Dixon, D. A., and Wiedner, E. S. (2024). Thermodynamic and kinetic activity descriptors for the catalytic hydrogenation of ketones. *J. Am. Chem. Soc.* 146, 6866–6879. doi:10.1021/jacs.3c13876

Das, P., Pan, S., and Chattaraj, P. K. (2023). "Planar hypercoordinate carbon," in *Atomic clusters with unusual structure, bonding and reactivity* (Elsevier), 357–372. doi:10.1016/B978-0-12-822943-9.00021-8

Dub, P. A., and Ikariya, T. (2012). Catalytic reductive transformations of carboxylic and carbonic acid derivatives using molecular hydrogen. *ACS Catal.* 2, 1718–1741. doi:10.1021/cs300341g

Dunne, J. F., Fulton, D. B., Ellern, A., and Sadow, A. D. (2010). Concerted C–N and C–H bond formation in a magnesium-catalyzed hydroamination. *J. Am. Chem. Soc.* 132, 17680–17683. doi:10.1021/ja108881s

Falconnet, A., Magre, M., Maity, B., Cavallo, L., and Rueping, M. (2019). Asymmetric magnesium-catalyzed hydroboration by metal-ligand cooperative catalysis. *Angew. Chem. Int. Ed.* 58, 17567–17571. doi:10.1002/anie.201908012

Frisch, M. J., Trucks, G. W., Schlegel, H. B., Scuseria, G. E., Robb, M. A., Cheeseman, J. R., et al. (2016). Gaussian 16, *revision B.01*. Wallingford, CT: Gaussian, Inc.

Garcia, L., Dinoi, C., Mahon, M. F., Maron, L., and Hill, M. S. (2019a). Magnesium hydride alkene insertion and catalytic hydrosilylation. *Chem. Sci.* 10, 8108–8118. doi:10.1039/C9SC02056J

Garcia, L., Mahon, M. F., and Hill, M. S. (2019b). Multimetallic alkaline-earth hydride cations. *Organometallics* 38, 3778–3785. doi:10.1021/acs.organomet.9b00493

Gonzalez, C., and Schlegel, H. B. (1989). An improved algorithm for reaction path following. *J. Chem. Phys.* 90, 2154–2161. doi:10.1063/1.456010

Harder, S. (2012). Molecular early main group metal hydrides: synthetic challenge, structures and applications. *Chem. Commun.* 48, 11165–11177. doi:10.1039/c2cc33478j

Hill, M. S., Liptrot, D. J., and Weetman, C. (2016). Alkaline earths as main group reagents in molecular catalysis. *Chem. Soc. Rev.* 45, 972–988. doi:10.1039/C5CS00880H

Inostroza, D., Leyva-Parra, L., Yañez, O., Solar-Encinas, J., Vásquez-Espinal, A., Valenzuela, M. L., et al. (2023). Searching for systems with planar Hexacoordinate carbons. *Atoms* 11, 56. doi:10.3390/atoms11030056

Jaszczewska-Adamczak, J. A., Baczewska, P., Bujok, R., and Mlynarski, J. (2024). Magnesium-catalyzed asymmetric thia-michael addition to α,β-Unsaturated ketones. *Adv. Synth. Catal.* 366, 1412–1421. doi:10.1002/adsc.202301414

Jin, B., Yan, M., Feng, L., Miao, C., and Wang, Y. (2024). $CBe_2H_5^-$: unprecedented 2σ/2π double aromaticity and dynamic structural fluxionality in a planar tetracoordinate carbon cluster. *Chem. - Eur. J.* 30, e202304134. doi:10.1002/chem.202304134

Johnson, N. B., Lennon, I. C., Moran, P. H., and Ramsden, J. A. (2007). Industrial-scale synthesis and applications of asymmetric hydrogenation catalysts. *Acc. Chem. Res.* 40, 1291–1299. doi:10.1021/ar700114k

Leyva-Parra, L., Diego, L., Yañez, O., Inostroza, D., Barroso, J., Vásquez-Espinal, A., et al. (2021). Planar hexacoordinate carbons: half covalent, half ionic. *Angew. Chem. Int. Ed.* 60, 8700–8704. doi:10.1002/anie.202100940

Li, X., Wang, L.-S., Boldyrev, A. I., and Simons, J. (1999). Tetracoordinated planar carbon in the Al_4C anion. A combined photoelectron spectroscopy and *ab initio* study. *J. Am. Chem. Soc.* 121, 6033–6038. doi:10.1021/ja9906204

Li, X., Zhang, H.-F., Wang, L.-S., Geske, G. D., and Boldyrev, A. I. (2000). Pentaatomic tetracoordinate planar carbon, $[CAl_4]^{2-}$: a new structural unit and its salt complexes. *Angew. Chem. Int. Ed.* 39, 3630–3632. doi:10.1002/1521-3773(20001016)39:20<3630::AID-ANIE3630>3.0.CO;2-R

Liang, Y., Das, U. K., Luo, J., Diskin-Posner, Y., Avram, L., and Milstein, D. (2022). Magnesium pincer complexes and their applications in catalytic semihydrogenation of alkynes and hydrogenation of alkenes: evidence for metal–ligand cooperation. *J. Am. Chem. Soc.* 144, 19115–19126. doi:10.1021/jacs.2c08491

Liptrot, D. J., Hill, P. M. S., and Mahon, M. F. (2014). Accessing the single-electron manifold: magnesium-mediated hydrogen release from silanes. *Angew. Chem. Int. Ed.* 53, 6224–6227. doi:10.1002/anie.201403208

Lu, T., and Chen, F. (2012). Multiwfn: a multifunctional wavefunction analyzer. *J. Comput. Chem.* 33, 580–592. doi:10.1002/jcc.22885

Magre, M., Maity, B., Falconnet, A., Cavallo, L., and Rueping, M. (2019). Magnesium-catalyzed hydroboration of terminal and internal alkynes. *Angew. Chem. Int. Ed.* 58, 7025–7029. doi:10.1002/anie.201902188

Magre, M., Szewczyk, M., and Rueping, M. (2022). s-Block metal catalysts for the hydroboration of unsaturated bonds. *Chem. Rev.* 122, 8261–8312. doi:10.1021/acs.chemrev.1c00641

Malhan, A. H., and Thirumoorthy, K. (2024). Global minimum and a heap of low-lying isomers with planar tetracoordinate carbon in the $CAl_3MgH_2^-$ system. *Phys. Chem. Chem. Phys.* 26, 3804–3809. doi:10.1039/D3CP05841G

Malhan, A. H., and Thirumoorthy, K. (2025). Catalytic hydrogenation of alkyne with planar tetracoordinate carbon in $CAl_3MgH_2^-$ system. *Front. Chem.* 13, 1672968. doi:10.3389/fchem.2025.1672968

Malhan, A. H., Sobinson, S., Job, N., Shajan, S., Mohanty, S. P., Thimmakondu, V. S., et al. (2022). $Al_2C_4H_2$ isomers with the planar tetracoordinate carbon (ptC)/Aluminum (ptAl). *Atoms* 10, 112. doi:10.3390/atoms10040112

Mawarnis, E. R., and Umar, A. A. (2019). Fibrous bimetallic silver palladium and ruthenium palladium nanocrystals exhibit an exceptionally high active catalytic process in acetone hydrogenation. *Mater. Today Chem.* 14, 100178. doi:10.1016/j.mtchem.2019.08.001

Monkhorst, H. J. (1968). "Activation energy for interconversion of enantiomers containing an asymmetric carbon atom without breaking bonds,"*Chem. Commun.* London, 1111–1112. doi:10.1039/c19680001111

Mukherjee, D., Ellern, A., and Sadow, A. D. (2014). Magnesium-catalyzed hydroboration of esters: evidence for a new zwitterionic mechanism. *Chem. Sci.* 5, 959–964. doi:10.1039/C3SC52793J

Naumkin, F. Y. (2008). Flat-structural motives in small alumino–carbon clusters C_nAl_m (n = 2–3, m = 2–8). *J. Phys. Chem. A* 112, 4660–4668. doi:10.1021/jp711230x

Neese, F. (2012). The ORCA program system. *WIREs Comput. Mol. Sci.* 2, 73–78. doi:10.1002/wcms.81

Nobel, N. K., Bamba, K., Patrice, O. W., and Ziao, N. (2017). NBO population analysis and electronic calculation of four azopyridine ruthenium complexes by DFT method. *Comput. Chem.* 05, 51–64. doi:10.4236/cc.2017.51005

Noyori, R., and Ohkuma, T. (2001). Asymmetric catalysis by architectural and functional molecular engineering: practical Chemo- and stereoselective hydrogenation of Ketones. *Angew. Chem. Int. Ed.* 40, 40–73. doi:10.1002/1521-3773(20010105)40:1<40::AID-ANIE40>3.0.CO;2-5

Papa, A. J. (2000). "Propanols," in *Ullmann's encyclopedia of industrial chemistry* (Weinheim, Germany: Wiley-VCH Verlag GmbH & Co. KGaA). doi:10.1002/14356007.a22_173

Rahman, A., and S-Al-Deyab, S. (2014). A review on reduction of acetone to isopropanol with Ni nano superactive, heterogeneous catalysts as an environmentally benevolent approach. *Appl. Catal. A* 469, 517–523. doi:10.1016/j.apcata.2013.10.015

Rauch, M., Ruccolo, S., and Parkin, G. (2017). Synthesis, structure, and reactivity of a terminal magnesium hydride compound with a carbatrane motif, $[Tism^{PriBenz}]MgH$: a multifunctional catalyst for hydrosilylation and hydroboration. *J. Am. Chem. Soc.* 139, 13264–13267. doi:10.1021/jacs.7b06719

Reed, A. E., Weinstock, R. B., and Weinhold, F. (1985). Natural population analysis. *J. Chem. Phys.* 83, 735–746. doi:10.1063/1.449486

Revunova, K., and Nikonov, G. I. (2015). Main group catalysed reduction of unsaturated bonds. *Dalton Trans.* 44, 840–866. doi:10.1039/C4DT02024C

Ried, A. C. A., Taylor, L. J., Geer, A. M., Williams, H. E. L., Lewis, W., Blake, A. J., et al. (2019). A highly active Bidentate magnesium catalyst for amine-borane dehydrocoupling: kinetic and mechanistic studies. *Chem. - Eur. J.* 25, 6840–6846. doi:10.1002/chem.201901197

Riplinger, C., and Neese, F. (2013). An efficient and near linear scaling pair natural orbital based local coupled cluster method. *J. Chem. Phys.* 138, 034106. doi:10.1063/1.4773581

Riplinger, C., Sandhoefer, B., Hansen, A., and Neese, F. (2013). Natural triple excitations in local coupled cluster calculations with pair natural orbitals. *J. Chem. Phys.* 139, 134101. doi:10.1063/1.4821834

Roy, M. M. D., Omaña, A. A., Wilson, A. S. S., Hill, M. S., Aldridge, S., and Rivard, E. (2021). Molecular main Group metal hydrides. *Chem. Rev.* 121, 12784–12965. doi:10.1021/acs.chemrev.1c00278

Rylander, P. N. (1979). *Catalytic hydrogenation in organic syntheses*. New York: Academic Press.

Sarmah, K., Purkayastha, S. K., and Guha, A. K. (2023). Reversible hydrogen storage by planar hypercoordinate carbon clusters. *Energy fuels.* 37, 9598–9609. doi:10.1021/acs.energyfuels.3c00486

Schlegel, H. B. (1982). Optimization of equilibrium geometries and transition structures. *J. Comput. Chem.* 3, 214–218. doi:10.1002/jcc.540030212

Shen, M., Zhao, G., Nie, Q., Meng, C., Sun, W., Si, J., et al. (2021). Ni-foam-structured Ni–Al_2O_3 ensemble as an efficient catalyst for gas-phase acetone hydrogenation to isopropanol. *ACS Appl. Mater. Interfaces* 13, 28334–28347. doi:10.1021/acsami.1c07084

Sun, R., Zhao, X.-F., Jin, B., Huo, B., Bian, J.-H., Guan, X.-L., et al. (2020). Influence of stepwise oxidation on the structure, stability, and properties of planar pentacoordinate carbon species CAl_5^+. *Phys. Chem. Chem. Phys.* 22, 17062–17067. doi:10.1039/D0CP01106A

Sun, R., Jin, B., Huo, B., Yuan, C., Zhai, H.-J., and Wu, Y.-B. (2022). Planar pentacoordinate carbon in a sulphur-surrounded boron wheel: the global minimum of $CB_5S_5^+$. *Chem. Commun.* 58, 2552–2555. doi:10.1039/D1CC07313C

Szewczyk, M., Magre, M., Zubar, V., and Rueping, M. (2019). Reduction of cyclic and linear organic carbonates using a readily available magnesium catalyst. *ACS Catal.* 9, 11634–11639. doi:10.1021/acscatal.9b04086

Thirumoorthy, K., Chandrasekaran, V., Cooksy, A. L., and Thimmakondu, V. S. (2020). Kinetic stability of $Si_2C_2H_2$ isomer with a planar tetracoordinate carbon atom. *Chem. East.* 3, 13–27. doi:10.3390/chemistry3010002

Tsuruoka, K., Koyasu, K., Hirabayashi, S., Ichihashi, M., and Tsukuda, T. (2018). Size-dependent polymorphism in aluminum carbide cluster anions $Al_nC_2^-$: formation of acetylide-containing structures. *J. Phys. Chem. C* 122, 8341–8347. doi:10.1021/acs.jpcc.7b12767

Wang, M., Kalita, A. J., Orozco-Ic, M., Yan, G., Chen, C., Yan, B., et al. (2023). Planar pentacoordinate s-block metals. *Chem. Sci.* 14, 8785–8791. doi:10.1039/D2SC05939H

Wang, L., Lv, J., Zhang, Y., and Yang, D. (2024). Asymmetric magnesium catalysis for important chiral scaffold synthesis. *Org. Biomol. Chem.* 22, 4778–4800. doi:10.1039/D4OB00521J

Weber, M., and Weber, M. (2010). "Phenols," in *Phenolic resins: a century of progress* (Berlin, Heidelberg: Springer Berlin Heidelberg), 9–23. doi:10.1007/978-3-642-04714-5_2

Weigend, F., and Ahlrichs, R. (2005). Balanced basis sets of split valence, triple zeta valence and quadruple zeta valence quality for H to Rn: design and assessment of accuracy. *Phys. Chem. Chem. Phys.* 7, 3297–3305. doi:10.1039/b508541a

Wirtz, L., Ghulam, K. Y., Morgenstern, B., and Schäfer, A. (2022). Constrained geometry *ansa* -half-sandwich complexes of magnesium – versatile *s* -block catalysts. *ChemCatChem* 14, e202201007. doi:10.1002/cctc.202201007

Wu, Y., Jiang, J., Lu, H., Wang, Z., Perez-Peralta, N., Islas, R., et al. (2011). Starlike aluminum–carbon aromatic species. *Chem. - Eur. J.* 17, 714–719. doi:10.1002/chem.201001266

Xu, J., Zhang, X., Yu, S., Ding, Y., and Bowen, K. H. (2017). Identifying the hydrogenated planar tetracoordinate carbon: a combined experimental and theoretical study of CAl_4H and CAl_4H^-. *J. Phys. Chem. Lett.* 8, 2263–2267. doi:10.1021/acs.jpclett.7b00732

Yang, L.-M., Ganz, E., Chen, Z., Wang, Z.-X., and Schleyer, P. V. R. (2015). Four decades of the chemistry of planar hypercoordinate compounds. *Angew. Chem. Int. Ed.* 54, 9468–9501. doi:10.1002/anie.201410407

Yang, Z., Li, D., Zhang, L., Zheng, J., Zhang, N., and Chen, B. H. (2017). Coupling synergetic effect between ruthenium and ruthenium oxide with size effect of ruthenium particles on ketone catalytic hydrogenation. *ChemCatChem* 9, 338–346. doi:10.1002/cctc.201601245

Yang, D., Wang, L., Li, D., and Wang, R. (2019). Magnesium catalysis in asymmetric synthesis. *Chem* 5, 1108–1166. doi:10.1016/j.chempr.2019.02.002

Ye, D., Jiang, M., Ning, L., Li, W.-Y., Lin, L., and Feng, X. (2025). Chiral Magnesium(II)-Catalyzed asymmetric hydroalkylation of imine-containing vinylazaarenes through conjugate addition. *Org. Lett.* 27, 3601–3606. doi:10.1021/acs.orglett.5c00668

Zhang, X., Emge, T. J., and Hultzsch, K. C. (2012). A chiral phenoxyamine magnesium catalyst for the enantioselective hydroamination/cyclization of aminoalkenes and intermolecular hydroamination of vinyl arenes. *Angew. Chem. Int. Ed.* 51, 394–398. doi:10.1002/anie.201105079

Zhang, C.-J., Wang, P., Xu, X.-L., Xu, H.-G., and Zheng, W.-J. (2021). Photoelectron spectroscopy and theoretical study of $Al_nC_5^{-/0}$ (n = 1–5) clusters: structural evolution, relative stability of star-like clusters, and planar tetracoordinate carbon structures. *Phys. Chem. Chem. Phys.* 23, 1967–1975. doi:10.1039/D0CP06081J

Zhang, C., Ortíz-Chi, F., Xu, X., Xu, H., Merino, G., and Zheng, W. (2023). Reconsidering the structures of $C_2Al_4^-$ and $C_2Al_5^-$. *Chem. - Eur. J.* 29, e202301338. doi:10.1002/chem.202301338

Zhang, M., Gu, J., and Zhao, Y. (2025). Au^I complexes with planar tetracoordinate carbon and their catalytic activity for the rearrangement of allylic acetates: a computational study. *ACS Omega* 10, 12514–12521. doi:10.1021/acsomega.5c00089

Zhao, Y., and Truhlar, D. G. (2008). The M06 suite of density functionals for main group thermochemistry, thermochemical kinetics, noncovalent interactions, excited states, and transition elements: two new functionals and systematic testing of four M06-class functionals and 12 other functionals. *Theor. Chem. Acc.* 120, 215–241. doi:10.1007/s00214-007-0310-x

Zhao, X.-F., Bian, J.-H., Huang, F., Yuan, C., Wang, Q., Liu, P., et al. (2018). Stabilization of beryllium-containing planar pentacoordinate carbon species through attaching hydrogen atoms. *RSC Adv.* 8, 36521–36526. doi:10.1039/C8RA07664B

Zheng, J., Xu, X., and Truhlar, D. G. (2011). Minimally augmented Karlsruhe basis sets. *Theor. Chem. Acc.* 128, 295–305. doi:10.1007/s00214-010-0846-z